Cross-Domain Deepfake Detection Based on Latent Domain Knowledge Distillation

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***Abstract*—The rapid development of deepfake technology poses challenges to face-centered data security. Existing methods primarily focus on how to transfer deepfake detectors from the source domain to the target domain to handle diverse deepfake techniques. In practical application scenarios, it is usually difficult to access the true and false labels of the source domain. In this letter, we introduce a new adaptation framework called Latent Domain Knowledge Distillation (LDKD) for cross-domain deepfake detection. In the proposed framework, we construct a knowledge distillation structure that includes a student network and a teacher network, which are jointly optimized in a coupled manner to facilitate the model's adaptation to the target domain. Furthermore, to improve the quality of pseudo-labels generated by the teacher network, we propose a Fourier Latent Domain Generation Module (FLGM) and a Stochastic Complementary Mask Module (SCMM). The former is used to generate latent domains to bridge domain differences at the image level, while the latter is employed to mine richer contextual cues for the model. Extensive cross-domain experimental results demonstrate that our method achieves state-of-the-art performance, and the model analysis proves the effectiveness of our key components.**

***Index Terms*—Cross-domain deepfake detection, Fourier latent domain generation, knowledge distillation, stochastic complementary mask.**

I. Introduction

THE rapid development of deep neural networks has led to significant advances in deepfake techniques, making it increasingly difficult to distinguish between real and fake content visually. The simplicity and accessibility of these techniques allow even ordinary users to create convincing deep forgeries, raising serious concerns about social security and personal privacy. Deepfake detection technology serves as an effective measure to counter deepfake techniques. A significant challenge in the field of deepfake detection is achieving satisfactory cross-domain performance on datasets that have not been previously encountered.

Recent studies have increasingly focused on enhancing the generalization ability of detectors, such as SLADD [1], SFFE [2]. Other research has explored leveraging information in the frequency domain for deepfake detection, including generating frequency-level perturbation maps for deepfake detectors [3], utilizing phase spectroscopy to identify upsampling artifacts in facial forgeries [4], and combining FFT with CNN classifiers [5]. When confronted with new forgery techniques, these methods require a large amount of labeled training data from new forgery methods to fine-tune detectors. However, they still face performance degradation when handling new test sets. Unsupervised Domain Adaptation (UDA) presents a viable solution. For the cross-domain problem of deepfake detection, most UDA methods [6], [7], [8] can reduce the difference between the distributions of the source and target domains at the image level or feature level, but these methods require source domain data for domain alignment. Besides, access to face source data may raise privacy concerns and create limitations. Compared with most methods that rely on labeled source data, pseudo-labeled methods [9], [10], [11] achieve Source-Free Domain Adaptation (SFDA), thereby alleviating privacy issues related to accessing source data. There are also methods [12], [13] that mitigate the difficulty of domain adaptation by using Knowledge Distillation (KD) techniques to assign pseudo-labels to the target domain.

Inspired by SFDA [9], we propose a new adaptation framework called Latent Domain Knowledge Distillation (LDKD). Firstly, we introduce a Fourier Latent Domain Generation Module (FLGM) to perform style transfer between unlabeled source and target domain images. Secondly, we introduce a Stochastic Complementary Mask Module (SCMM) to generate stochastic complementary mask images, encouraging the model to learn contextual relationships and reduce domain differences. A pair of teacher-student models is also included in our framework, with pseudo-labels generated by the teacher model. The latent images generated by FLGM serve as a bridge between the source and target domains, helping the teacher model improve the quality of generating pseudo-labels. SCMM encourages model learning of contextual relationships, thereby reducing domain differences. The contributions of this letter can be concluded as follows:

1) We propose a LDKD framework for cross-domain deepfake detection. Our LDKD leverages knowledge distillation to facilitate deepfake detection in a novel approach.

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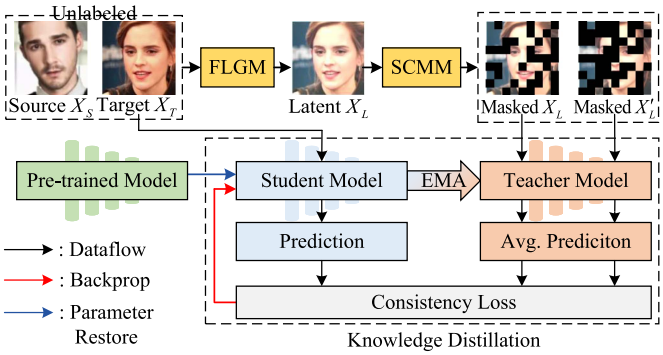


Fig. 1. Overview of the proposed LDKD framework. We first use the FLGM to reduce the domain difference. Complementary mask images are then generated using the SCMM. The complementary mask images are fed into the teacher model, which generates pseudo-labels to guide the student model in better adapting to the target domain.

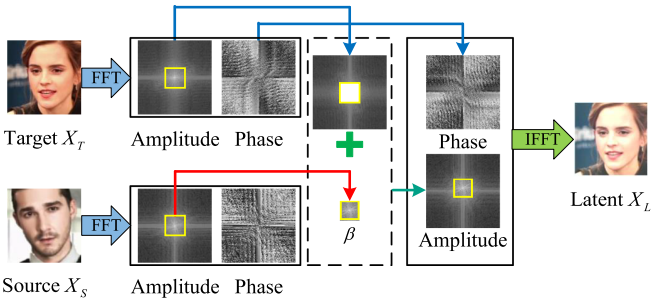


Fig. 2. During the testing phase, we use the FLGM to generate latent images that merge the content of target images with the specific style of the source image. This is achieved by replacing the low-frequency components of the target image's Fourier magnitude with those from the source image.

2) We propose a FLGM to achieve online style transfer. Furthermore, we introduce a SCMM to enhance the model's ability to capture richer contextual cues.
3) The proposed method achieves superior performance compared to the state-of-the-art methods in both within-dataset and across-dataset experiments. Ablation study also validates the effectiveness of each component in LDKD.

II. Proposed Method

A. Overview of the LDKD Framework

To improve the generalization of the model, we propose a generic framework named LDKD and construct a knowledge distillation structure in the framework that includes a student network and a teacher network, which are jointly optimized in a coupled manner [14] to facilitate the model's adaptation to the target domain. Furthermore, to enhance the quality of the pseudo-labels generated by the teacher network, we propose a Fourier Latent Domain Generation Module (FLGM) and a Stochastic Complementary Mask Module (SCMM), as shown in Fig. 1. Firstly, we use the FLGM to generate latent images containing the style of source images and the content of target images, and extract richer cross-domain knowledge from the joint distribution space. Subsequently, the generated latent domain images are fed into the SCMM to generate two complementary masked latent domain images, enhancing the model's ability to learn the contextual content of the local region. Finally, complementary masked latent domain images are fed into the teacher model to help knowledge extraction from the teacher model to the student model, improving the performance of the student model.

B. Fourier Latent Domain Generation Module

For domain adaptation deepfake detection, we define the source domain, target domain, and latent domain as $\mathcal{D}_S = \{X_S^i, Y_S^i\}_{i=1}^{N_S}$, $\mathcal{D}_T = \{X_T^i\}_{i=1}^{N_T}$, and $\mathcal{D}_L = \{X_L^i\}_{i=1}^{N_L}$, respectively. Given a pre-trained model f_θ, the parameters θ are trained on a given unlabeled dataset $\mathcal{D}_S$. Our FLGM uses a non-adversarial photometric alignment method. Despite its simplicity, it effectively removes perturbations from the data and learns domain-invariant representations by minimizing the distribution differences between the source and target domains. In comparison to other domain adaptation methods [15] that rely on feature-level and image-level alignment, our Fourier transform-based method offers a cost-effective and streamlined solution to achieve domain alignment.

In the Fourier frequency domain, the low-frequency amplitude components, i.e. the central region of the amplitude spectrum, carry more domain-specific information. This approach not only effectively eliminates the domain discrepancies, but also preserves the semantic information of the image, thus improving the generalization performance of the cross-domain scenarios. Given a single image $x \in \mathbb{R}^{H\times W\times 3}$, its Fourier transform $\mathcal{F}$ is:

$$\mathcal{F}(x)(m,n) = \sum_{h,w} x(h,w) e^{-j2\pi\left(\frac{h}{H}m + \frac{w}{W}n\right)}, j^2 = -1. \quad (1)$$

The FFT algorithm [16] is used to efficiently calculate the Fourier transform. Further, a mask M_β is designed to represent the low-frequency mask, which has zero value except for the central region $\beta \in (0, 1)$:

$$M_\beta(h,w) = \mathbb{I}_{(h,w)} \in [-\beta H : \beta H, -\beta W : \beta W], \quad (2)$$

where the center of the image is the coordinate (0, 0). The Fourier latent domain image can be obtained by (3). First, the low-frequency amplitude of the source image X_T is replaced with that of the target image X_S. Next, the modified spectral data X_T, with its phase preserved, is inverted to produce the latent image X_L. This latent image X_L retains the content of X_T while adopting the style of X_S. This process is illustrated in Fig. 2, where M_β is shown in yellow.

$$X_L = \mathcal{F}^{-1}([M_\beta \circ \mathcal{F}^A(X_S) + (1 - M_\beta) \circ \mathcal{F}^A(X_T), \mathcal{F}^P(X_T)]), \quad (3)$$

where $\mathcal{F}^{-1}(\cdot)$ denotes inverse Fourier transform, "$\circ$" denotes the element dot product operation, $\mathcal{F}^A$ and $\mathcal{F}^P$ respectively denote the magnitude and phase components of the Fourier transform.

C. Stochastic Complementary Mask Module

Masked image modeling has emerged as a distinctive approach in self-supervised learning, where portions of the original data are proportionally masked and then predicted. This paradigm allows deep models to learn robust representations, leading to superior performance across various tasks, including

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object localization [17], self-supervised feature learning [18] and semantic segmentation [19].

Unlike traditional masking techniques that merely obscure a part of an image, our SCMM enhances the training process by generating two complementary masked latent images. Despite both images being partially obscured by the mask, they together provide a complete image, with the missing components in one being present in the other. This strategy not only enhances data variety but also strengthens the model's generalization ability.

We preserve local information by stochastically masking out parts of the target image. To achieve this, a patch mask M is randomly sampled from a uniform distribution:

$$M_{\substack{ap+1:(a+1)p,\\ bp+1:(b+1)p}} = \begin{cases} 1 & \text{if } v > r, \\ 0 & \text{if } v \le r, \end{cases} \quad \text{with } v \sim \mathcal{U}(0,1), \tag{4}$$

where p is the patch size, r is the mask ratio, and a, b are the patch indices. The latent domain image X_L is fed into our SCMM. In this process, one-half of X_L is randomly obscured by an element-wise multiplication with the mask M, resulting in a partially concealed image. Subsequently, X_L undergoes a second masking operation in a complementary fashion, yielding two images that complete each other in the latent domain, $\tilde{X}_L$ and $\tilde{X}'_L$. The complementary stochastic masking process is delineated as follows:

$$\begin{cases} \tilde{X}_L = M \odot X_L \\ \tilde{X}'_L = (1 - M) \odot X_L \end{cases}, \tag{5}$$

where $\odot$ denotes element-by-element multiplication.

D. Knowledge Distillation

To adapt a pre-trained model for use in an unlabeled target domain, the majority of domain adaptation algorithms rely on the use of pseudo-labels during the testing process. Obtaining reliable pseudo-labels in this process is of utmost importance. Therefore, we filter the pseudo-labels by setting a confidence threshold, where the difference in confidence reflects the domain difference. The teacher model, denoted as f_{θ_T}, is initialized using the parameters of the pre-trained model f_θ. The teacher model generates pseudo-labels for the unlabeled input images X_T, the direct predictions of high confidence inputs by the teacher model are as follows:

$$y'_T = f_{\theta_T}(X_T), \tag{6}$$

where y'_T is the prediction of the teacher model for input above a threshold. When the input is below the confidence threshold, we combine it with FLGM and SCMM, it is defined as:

$$\hat{y}_T = f_{\theta_T}(\alpha_i(\tilde{X}_L)), \tag{7}$$

$$\hat{y}'_T = f_{\theta_T}(\alpha_i(\tilde{X}'_L)), \tag{8}$$

where $\alpha_i()$ is the data augmentation. We refine the student model by leveraging a consistency cross-entropy loss, which makes the teacher and student predictions tend to be consistent:

$$\mathcal{L}_{total} = \mathcal{L}_{ce}(\hat{y}_S, \hat{y}_T) + \lambda \mathcal{L}_{ce}(\hat{y}_S, \hat{y}'_T), \tag{9}$$

where $\mathcal{L}_{ce}$ refers to the consistency cross-entropy loss, and $\hat{y}_S$ is the student model predictions. $\mathcal{L}_{total}$ calculate the predictions of the student model and teacher model of the input complementary mask images.

TABLE I
THE INFLUENCE OF DIFFERENT PATCH SIZES

Patch	AUC		
	DF	FS	NT
16	87.5	63.8	63.0
32	**89.3**	**66.2**	**67.7**
64	88.7	65.9	67.4
128	88.1	65.7	66.8

TABLE II
ABLATION STUDY ON THE INFLUENCE OF DIFFERENT COMPONENTS

ID	AUC		
	DF	FS	NT
(a)	86.5	51.7	54.4
(b)	88.6	65.9	67.1
(c)	87.9	64.7	66.3
LDKD	**89.3**	**66.2**	**67.7**

To optimize the teacher model, the parameters of the teacher network are updated by an Exponential Moving Average (EMA) [20]:

$$\theta_S^{t+1} = \kappa \theta_S^t + (1 - \kappa)\theta_T^t, \tag{10}$$

where κ is a smoothing factor, t denotes the training step. The teacher model enables the temporal ensemble of the student model, increasing the quality and temporal stability of the pseudo-labels. Although high-quality pseudo-labels can help reduce error accumulation, new errors are inevitably introduced, so we randomly restore a small fraction of neurons to the pre-training source parameters at each iteration to avoid catastrophic forgetting. θ_0^f denotes the initial parameters of the source domain update, and the parameters of the student network updated by random recovery can be expressed as:

$$\psi \sim Bernoulli(\rho), \tag{11}$$

$$\theta_S^{t+1} = \psi \odot \theta_0^f + (1 - \psi) \odot \theta_S^{t+1}, \tag{12}$$

where $\odot$ denotes the elemental multiplication, ρ is the stochastic recovery probability, and ψ denotes the partial recovery of parameters within the mask θ_S^{t+1} to the initial weights θ_0^f. With stochastic recovery, the network avoids deviating too far from the initial source model, thus avoiding the catastrophic forgetting problem caused by error accumulation.

III. EXPERIMENTS

A. Experimental Settings

1) Datasets: We evaluate our proposed method on two challenging datasets, including FaceForensics++ (FF++) [21] and CelebDF [22]. FF++ dataset consists of a total of 1,000 real videos (720 videos for training, 140 videos for validation, and 140 videos for testing) and 4,000 fake videos generated from the real videos using four different forgery methods, i.e., FaceSwap (FS), Face2Face (F2F), DeepFakes (DF), and NeuralTextures (NT). Each video in FF++ is available in three quality levels:

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unaltered raw format, lightly compressed (HQ), and heavily compressed (LQ). We adopt the HQ version by default and specify the version otherwise. To fully evaluate the generalization, we use the Celeb-DF-v2 dataset, which provides 590 real videos and 5,639 fake videos. The fake facial videos are generated by improved deepfake synthesis techniques, where those fake faces are more realistic and thus harder to distinguish for human eyes.

2) Implementation Details: Our proposed method is executed within the PyTorch framework, utilizing two NVIDIA GTX 3090 GPUs. We have adapted the Xception architecture [23] to serve as the backbone of our network. The model's parameters are initialized by Xception pre-trained on ImageNet [23]. For facial extraction and alignment, we employ the DLIB library, and subsequently resized the aligned facial images to 299×299 pixels. The batch size for training is configured at 32. The Adam optimizer is engaged in the optimization process, with a learning rate of 0.0002. Furthermore, the parameters b and r in (4) are set to 32 and 0.5, respectively.

3) Evaluation: Following the standard setting, we use the Area Under Receiver (AUC) as the evaluation metric, which evaluates the model's capacity to distinguish between real and fake faces. To demonstrate the effectiveness of our approach, we compare it with several state-of-the-art methods: CORE [25], MAT [26], LTW [27], CADDM [28], UCF [29], DFIL [30], AF [31], OST [32], FDA [33] and Face X-ray [34].

B. Ablation Study

Influence of p: Table I shows the study of patch block size selection. We use F2F for training and use three other methods for testing. The best performance is achieved at $p = 32$ for a fixed value of $r = 0.5$.

Influence of proposed components: We conduct the ablation study on different components proposed in our framework to evaluate their effectiveness. We use F2F for training and use three other methods for testing. The results are summarized in Table II. We individually evaluated each part: (a) the baseline Xception, (b) the proposed method without SCMM, and (c) the proposed method without FLGM.

C. Comparison With State-of-the-Art Methods

1) Comparison Results on FF++: As shown in Table III, our LDKD framework demonstrates superior performance in most cases when compared to other methods. We use forged images from one method for training and those from three other methods for testing. Our approach enhances the model's generalization by fully utilizing the information in the latent domain and the contextual content of the complementary mask, thus improving its ability to generalize from one method to another.

2) Cross-Dataset Evaluation on Celeb-DF-V2: Table IV presents the AUC comparison with recent deepfake detection methods. Our method achieves a leading AUC on Celeb-DF-v2, demonstrating the effectiveness of our LDKD in cross-dataset deepfake detection.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS

Train	Method	DF	F2F	FS	NT	Avg
DF	Xception [23]	–	69.4	39.7	64.9	58.0
	CORE [25]	–	67.6	38.2	75.1	60.3
	MAT [26]	–	74.2	39.5	68.8	60.8
	LTW [27]	–	74.4	46.3	71.9	64.2
	CADDM [28]	–	69.9	**53.8**	62.9	62.2
	UCF [29]	–	69.4	49.2	64.8	61.1
	DFIL [30]	–	73.8	50.3	75.3	66.5
	AF [31]	–	74.1	42.1	74.8	63.7
	Ours	**–**	**74.6**	52.2	**76.1**	**67.6**
F2F	Xception [23]	86.5	–	51.7	54.4	64.2
	CORE [25]	73.9	–	53.6	49.0	58.8
	MAT [26]	85.2	–	60.8	63.6	69.9
	LTW [27]	88.9	–	61.3	63.7	71.3
	CADDM [28]	75.1	–	55.4	66.1	65.5
	UCF [29]	75.9	–	52.3	58.4	62.2
	DFIL [30]	80.3	–	59.5	61.0	66.9
	AF [31]	83.8	–	51.1	**75.8**	70.2
	Ours	**89.3**	**–**	**66.2**	67.7	**74.4**
FS	Xception [23]	51.4	58.0	–	48.5	52.6
	CORE [25]	66.1	63.3	–	54.3	61.2
	MAT [26]	64.1	**66.4**	–	49.1	59.9
	LTW [27]	68.4	64.7	–	46.2	59.8
	CADDM [28]	54.1	58.1	–	57.4	56.5
	UCF [29]	67.1	60.5	–	53.5	60.3
	DFIL [30]	52.7	64.6	–	46.7	54.7
	AF [31]	63.4	58.2	–	62.7	61.4
	Ours	**73.0**	64.9	**–**	**71.2**	**69.7**
NT	Xception [23]	83.8	67.7	43.3	–	64.9
	CORE [25]	83.6	69.3	47.4	–	66.8
	MAT [26]	85.4	52.2	76.3	–	71.3
	LTW [27]	87.4	50.5	72.6	–	70.2
	CADDM [28]	86.2	75.1	65.2	–	75.5
	UCF [29]	77.9	67.6	52.8	–	66.1
	DFIL [30]	**94.2**	**78.0**	51.0	–	74.4
	AF [31]	86.3	77.1	53.8	–	72.4
	Ours	90.0	77.3	**78.3**	**–**	**81.9**

Bold indicates the best results, and underline indicates the second best results.

TABLE IV
CROSS-DATASET EVALUATION FROM FF++ TO CELEB-DF-V2 IN TERMS OF AUC

Train	Method	Celeb-DF-v2
FF++	Xception [23]	68.6
	LTW [27]	64.1
	CADDM [28]	71.8
	OST [32]	74.8
	FDA [33]	70.1
	Face X-ray [34]	74.8
	Ours	**76.9**

IV. CONCLUSION

In this letter, we propose a LDKD framework for domain alignment using latent domain and a stochastic complementary masking strategy. The latent images generated by FLGM help extract rich cross-domain knowledge. The complementary images generated by SCMM are used to adjust the model, increasing its ability to learn contextual information. Experimental results show that the proposed method is effective for deepfake detection both within-domain and cross-domain. Considering the requirements of real-time applications, future work will focus on optimizing the speed of cross-domain deepfake detection.

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